Module 4 Assignment: Structure-Based Druggability Assessment

Course: BIOT 6900 – AI-Powered Biotech Workflows
Disease: Lung Adenocarcinoma (LUAD)
Targets: SCN7A and MASP1

1. Structural Confidence

For this assignment, I selected SCN7A and MASP1 from my Module 2 target list. I used AlphaFold structures to check their confidence and identify possible drug-binding pockets.

SCN7A had a mean pLDDT of 73.74, with 14.74% of its residues below 50. The main low-confidence regions were 437–473, 770–914, and 1646–1682. This means some parts of the structure may be flexible or not predicted accurately.

MASP1 had a higher mean pLDDT of 90.46, with only 2.72% of residues below 50. Its main low-confidence region was residues 1–18.

From the PAE analysis, SCN7A showed two confident regions with a mean inter-domain PAE of 20.72 Å. This means the relative positioning of these regions is uncertain. MASP1 showed one connected confident region in this analysis.

According to the lecture, regions with high pLDDT and low PAE are more reliable for pocket analysis, while low-confidence regions should be avoided (Sudhakaran, 2026).

Based on these results, I would trust the confident regions of MASP1 more. For SCN7A, I would focus only on regions with good confidence and low PAE.

2. Validation with Experimental Structures

I compared both AlphaFold models with their experimental PDB structures to check how closely they matched.

Target	Experimental PDB	Residues compared	RMSD
SCN7A	7TJ8, chain A	954	5.35 Å
MASP1	4IGD, chain A	377	1.68 Å

SCN7A had an RMSD of 5.35 Å, showing some differences between the predicted and experimental structures. However, the RMSD for its first detected domain was 3.37 Å. The largest differences were around residues 1635, 1632, and 762.

MASP1 had a lower RMSD of 1.68 Å, showing better agreement in the compared region. The largest differences were around residues 676 and 669.

These differences can happen because the experimental structures and AlphaFold models may represent different conditions. The experimental MASP1 structure also contains a modified fragment rather than the complete protein.

p53 anchor check

I also compared the AlphaFold p53 model with the experimental structure 1TUP, chain B.

The RMSD was 0.68 Å across 192 confident residues, showing very good agreement.

The residues C176, H179, C238, and C242 were close to zinc, while R248 and R273 were close to DNA at 3.4 Å and 2.5 Å, respectively.

In Module 3, the R175H variant had a score of -5.97. However, R175 was not directly contacting zinc or DNA in the structure. This shows that sequence scores alone cannot fully explain how a mutation affects a protein.

Overall, comparing predicted structures with experimental structures helps us identify which regions are more reliable.

3. Pocket Analysis and Druggability

I used fpocket to identify possible drug-binding pockets in both proteins. I followed the assignment rules: reject pockets with mean pLDDT below 70, mark pockets with PAE above 10 Å as suspect, and keep the remaining pockets.

SCN7A results

Pocket	Druggability score	Mean pLDDT	PAE (Å)	Decision
98	0.982	85.06	3.88	Keep
131	0.965	87.90	2.82	Keep
115	0.942	88.19	2.46	Keep
149	0.926	64.20	8.13	Reject
45	0.727	77.99	4.27	Keep

Pocket 98 had the highest score of 0.982. It also had good structural confidence and low PAE. Pocket 149 had a high score but was rejected because its mean pLDDT was below 70.

MASP1 results

Pocket	Druggability score	Mean pLDDT	PAE (Å)	Decision
1	0.687	70.91	5.89	Keep
16	0.190	93.44	1.66	Keep
21	0.152	90.25	1.43	Keep
2	0.077	93.86	1.87	Keep

7	0.033	92.09	1.25	Keep

For MASP1, pocket 1 had the highest score of 0.687. It passed the confidence checks, but its mean pLDDT was only slightly above 70.

Based on these results, my druggability call for SCN7A was **"Structure supports a small-molecule pocket"**, while for MASP1 it was **"Possible but uncertain."**

A high pocket score does not confirm that a drug will bind. It only suggests that the pocket has suitable features for binding (Le Guilloux et al., 2009).

4. Integration with Modules 2 and 3

In Module 2, SCN7A ranked 5th and MASP1 ranked 11th. After adding the Module 3 tractability scores, SCN7A moved to rank 3 and MASP1 moved to rank 8.

SCN7A had a small-molecule tractability score of 0.645 and an antibody score close to zero. MASP1 had scores of 0.605 for small molecules and 0.550 for antibodies.

In Module 4, SCN7A showed a high-scoring pocket of 0.982, while MASP1 had a lower best pocket score of 0.687.

These results gave me additional structural evidence to compare both targets.

I would prioritize SCN7A for further small-molecule studies because its best pocket had a higher druggability score and passed the confidence checks.

I would keep MASP1 as a secondary target because it also has a possible pocket, but further testing is needed.

5. Limitations and Next Steps

One limitation of AlphaFold is that it predicts one structure, but proteins can move and change shape. Some pockets may only become visible when a protein moves or when a molecule binds.

Another limitation is that AlphaFold models may not include metals, cofactors, DNA, binding partners, or the membrane environment. This is important for SCN7A because it is a membrane protein. MASP1 can also interact with other proteins, which may affect its structure.

AlphaFold is also not designed to predict the effects of individual mutations or new pockets formed by mutations (Sudhakaran, 2026).

For SCN7A, I would test possible small molecules using binding studies and ion-channel activity assays.

For MASP1, I would compare its active and inactive structures and test possible inhibitors using enzyme activity assays.

These experiments would help confirm whether the predicted pockets are useful drug-binding sites.

6. Ligand-Site Check

For SCN7A, I compared the predicted pockets with the ligand-contact regions found in experimental structure 7TJ8.

The best kept pocket was pocket 98, with a druggability score of 0.982 and rank 1.

This pocket covered 42% of the contact residues for PEV A2401.

However, another PEV molecule, A2405, matched pocket 86 more closely, with 62% coverage. Pocket 86 ranked 26th and had a low score of 0.014.

For PEV A2413, pocket 131 gave 58% coverage and ranked 2nd with a score of 0.965.

This shows that the highest-scoring pocket does not always match every ligand-binding region. Some real binding sites may have low fpocket scores because they are open, flexible, or located near the membrane rather than forming a deep pocket.

The molecules identified here are mainly associated with the membrane environment. Therefore, the results do not prove that these pockets bind therapeutic drugs.

For MASP1, experimental structure 4IGD did not contain a qualifying bound small-molecule ligand for comparison.

Overall, this check shows why pocket scores should be compared with experimental evidence before deciding whether a protein is druggable.

References

Le Guilloux, V., Schmidtke, P., & Tuffery, P. (2009). Fpocket: An open source platform for ligand pocket detection. *BMC Bioinformatics, 10*, Article 168. https://doi.org/10.1186/1471-2105-10-168

Sudhakaran. (2026). *Module 4: Structure prediction and structure-based druggability* [Course lecture slides]. Northeastern University.